

Table 1. **The robot-evaluation benchmark landscape: 160 web-verified benchmarks, compared on four axes.** Grouped into four evaluation lanes and sub-grouped by capability. Columns: **Year**; **Capability** (what the benchmark stresses); **Eval** (evaluation mode: open-loop prediction/generation quality, closed-loop task success, or a prediction-to-action bridge); and **VLA vs WM** (does the benchmark build a native direct-policy vs world-model contrast: ✓ explicit, ~ partial, -- model-agnostic). Every entry is a real, individually citeable benchmark.

Benchmark	Year	Capability	Eval	VLA vs WM
Policy / manipulation suites (37)				
<i>Manipulation</i>				
ARNOLD [55]	2023	language-grounded continuous-state 3D manipulation	closed	--
BiGym [32]	2024	mobile bimanual manipulation	closed	--
FMB [90]	2024	real-world functional multi-step assembly manipulation	closed	--
Franka Kitchen [60]	2019	multi-task kitchen manipulation	closed	--
FurnitureBench [63]	2023	real-world furniture assembly	closed	--
GRUtopia [133]	2024	city-scale embodied navigation and manipulation	closed	--
KitchenShift	2021	kitchen distribution-shift generalization	closed	--
ManiSkill2 [58]	2023	generalizable skills	closed	--
PerAct [120]	2022	language-conditioned 6-DoF single-arm manipulation	closed	--
PerAct2 [56]	2024	bimanual language-conditioned manipulation	closed	--
Ravens [145]	2020	vision-based pick-and-place rearrangement	closed	--
RGB-Stacking [79]	2021	vision-based stacking of diverse shapes	closed	--
RLBench [66]	2020	manipulation breadth	closed	--
RoboAgent [13]	2023	multi-task manipulation with RoboSet	closed	--
robomimic [93]	2021	single-arm manipulation from human demonstrations	closed	--
robosuite [155]	2020	modular simulated single-arm manipulation tasks	closed	--
RoboTwin [99]	2024	dual-arm bimanual manipulation	closed	--
VLMbench [153]	2022	vision-and-language compositional manipulation	closed	--
<i>Generalization & robustness</i>				
GemBench [53]	2024	generalization levels	closed	--
LIBERO [87]	2023	short-horizon + transfer	closed	--
RoboArena [4]	2025	real-world generalization	closed	--
SIMPLER / SimplerEnv [84]	2024	generalization (real vs sim)	closed	--
THE COLOSSEUM [107]	2024	generalization / robustness	closed	--
VIMA-Bench [72]	2023	multimodal-prompt generalization	closed	--
<i>Other</i>				
CortexBench [91]	2023	pretrained visual representations for embodied control	closed	--
GenManip [49]	2025	LLM-scene tasks	closed	~
HandoverSim [22]	2022	human-to-robot object handover	closed	--
Meta-World [144]	2020	multi-task / meta-RL	closed	--
RoboHive [78]	2023	unified robot-learning environments	closed	--
<i>Dexterous manipulation</i>				
Bi-DexHands [28]	2022	bimanual dexterous manipulation	closed	--
DeformableGym	2023	3D deformable-object grasping	closed	--
DexArt [9]	2023	dexterous articulated-object manipulation	closed	--
UniDexGrasp [138]	2023	universal dexterous grasping	closed	--
<i>Long-horizon & planning</i>				
CALVIN [96]	2021	long-horizon language	closed	--
RoboCasa [100]	2024	long-horizon (data scaling)	closed	--
VLABench [151]	2025	long-horizon reasoning	closed	--
<i>Deformable-object manipulation</i>				
SoftGym [86]	2020	deformable-object manipulation	closed	--
Embodied navigation, planning & social (85)				
<i>Long-horizon & planning</i>				

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
Alexa Arena [50]	2023	task-planning	closed	--
ALFRED [121]	2019	long-horizon, partial-obs	closed	--
BEHAVIOR-1K [81]	2024	long-horizon household	closed	--
CoELA (C-WAH/TDW-MAT) [147]	2023	long-horizon	closed	--
DialFRED [52]	2022	task-planning	closed	--
EgoPlan-Bench [27]	2023	task-planning	closed	--
EmbodiedBench [139]	2025	long-horizon	closed	--
EmbodiedCity [48]	2024	long-horizon	closed	--
EmbodiedEval [30]	2025	long-horizon	closed	--
HomeRobot OVMM [141]	2023	long-horizon	closed	--
LEGENT [31]	2024	long-horizon	closed	--
LH-VLN [124]	2024	long-horizon	closed	--
LoHoRavens [150]	2023	long-horizon	closed	--
LoTa-Bench [34]	2024	task-planning	closed	--
MFE-ETP [149]	2024	task-planning	closed	--
PARTNR [21]	2024	task-planning	closed	--
PlanBench [132]	2022	task-planning	closed	--
SafeAgentBench [143]	2024	task-planning	closed	--
SMART-LLM [73]	2023	task-planning	closed	--
TDW-Transport [46]	2021	task-planning	closed	--
TidyBot [135]	2023	task-planning	closed	--
VirtualHome [104]	2018	task-planning	closed	--
Watch-And-Help [105]	2020	long-horizon	closed	--
Social & multi-agent				
CrowdNav [24]	2018	social	closed	--
Habitat 3.0 [106]	2023	social / multi-agent	closed	--
HuNavSim [103]	2023	social	closed	--
JRDB [94]	2019	social	closed	--
JRDB-Act [42]	2021	social	closed	--
Melting Pot [80]	2021	social	closed	--
Overcooked-AI [19]	2019	social	closed	--
RoboSense [126]	2024	social	closed	--
SEAN [130]	2020	social	closed	--
SMAC [114]	2019	social	closed	--
SocialGym 2.0 [125]	2023	social	closed	--
SocNavBench [14]	2021	social navigation	closed	--
Navigation				
Gibson Env [137]	2018	navigation	closed	--
GOAT-Bench [74]	2024	multi-modal lifelong navigation	closed	--
Habitat [115]	2019	navigation	closed	--
Habitat 2.0 [129]	2021	rearrangement	closed	--
HM3D [111]	2021	navigation	closed	--
iGibson [119]	2020	interactive navigation	closed	--
Matterport3D [20]	2017	navigation	closed	--
ObjectNav [11]	2020	object-goal navigation	closed	--
ProcTHOR [39]	2022	navigation	closed	--
RoboTHOR [38]	2020	navigation / sim-to-real	closed	--
SoundSpaces [23]	2020	audio-visual navigation	closed	--
Autonomous driving				
Bench2Drive [69]	2024	driving	closed	--
CARLA [40]	2017	driving	closed	--
DriveArena [140]	2024	driving	closed	✓
MetaDrive [83]	2021	driving	closed	--
NAVSIM [37]	2024	driving	closed	--
nuScenes [17]	2019	driving	closed	--
Waymax [59]	2023	driving	closed	--
Waymo Open [128]	2019	driving	closed	--
Counterfactual reasoning				
ACRE [146]	2021	counterfactual	open	✓
CausalCity [95]	2021	counterfactual	closed	✗
CausalVQA [44]	2025	counterfactual	open	✓

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
CausalWorld [1]	2020	counterfactual	closed	✓
CoPhy [10]	2019	counterfactual	open	✓
CRAFT [3]	2020	counterfactual	open	✓
Filtered-CoPhy [67]	2022	counterfactual	open	✓
<i>Locomotion</i>				
Barkour [18]	2023	legged locomotion	closed	--
Brax [45]	2021	locomotion	closed	--
DERL [61]	2021	locomotion	closed	--
HumanoidBench [117]	2024	humanoid locomotion	closed	--
Isaac Gym [92]	2021	locomotion	closed	--
Legged Gym [113]	2021	legged locomotion	closed	--
<i>Theory of mind</i>				
AGENT [123]	2021	theory-of-mind	open	✓
BIB [47]	2021	theory-of-mind	open	✓
PHASE [101]	2021	theory-of-mind	closed	~
SocialAI [75]	2021	theory-of-mind	closed	--
ToMi	2019	theory-of-mind	closed	~
<i>Vision-language navigation</i>				
R2R [2]	2018	VLN	closed	--
REVERIE [108]	2020	VLN	closed	--
RxR [77]	2020	VLN	closed	--
Touchdown [26]	2019	outdoor VLN	closed	--
VLN-CE [76]	2020	VLN	closed	--
<i>Hidden-state / object permanence</i>				
Bongard-HOI [71]	2022	hidden-state	open	~
ComPhy [29]	2022	hidden-state	open	✓
LIBERO-Mem [36]	2025	hidden-state / occlusion	closed	✓
TEAch [102]	2021	hidden-state (dialog)	closed	--
<i>Embodied question answering</i>				
OpenEQA	2024	EQA	closed	--
RoboVQA [116]	2023	EQA	closed	--
<i>Generalization & robustness</i>				
ALFWorld [122]	2020	abstract-vs-grounded transfer	closed	~
<i>Other</i>				
O-PIAAGETS	2022	object permanence	closed	~
World-model & video-generation evaluation (34)				
<i>Generation quality</i>				
CATER [54]	2019	compositional-action temporal reasoning	open	--
DEVIL [85]	2024	content-dynamics quality of T2V models	open	--
EVA-Bench [33]	2024	embodied video anticipation	open	--
EvalCrafter [88]	2023	generation quality (17-metric suite)	open	--
EWMBench [64]	2025	scene/motion/semantic quality	open	--
FETV [89]	2023	fine-grained text-to-video quality	open	--
IPV-Bench [5]	2025	impossible-video generation + understanding	open	--
StoryBench [16]	2023	continuous story visualization	open	--
T2V-CompBench [127]	2024	compositional text-to-video quality	open	--
T2VScore [136]	2024	text-to-video metric (alignment + quality)	open	--
TC-Bench [43]	2024	temporal compositionality in video generation	open	--
VBench [65]	2023	generation quality (16 disentangled dimensions)	open	--
VBench-2.0 [152]	2025	generation faithfulness	open	--
VideoCon [6]	2023	robust video-language alignment	open	--
VideoHalluciner [134]	2024	hallucination in video-language models	open	--
VideoScore [62]	2024	learned automatic quality metric	open	--
WorldModelBench [82]	2025	video-WM prediction quality	open	--
WorldScore [41]	2025	world-gen quality	open	--
<i>Physical reasoning</i>				
ContPhy [154]	2024	continuum (soft-body/fluid) physical reasoning	open	--
GRASP [68]	2023	grounding + intuitive physics in video MLLMs	open	--
IntPhys [112]	2018	intuitive physics (violation-of-expectation)	open	--
IntPhys 2 [15]	2025	intuitive physics in complex scenes	open	--
PHYBench [110]	2025	physical perception and reasoning	open	--

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
PhyGenBench [97]	2024	physical-commonsense correctness in video gen	open	--
PhysBench [35]	2025	physical-world understanding for VLMs	open	--
Physics-IQ [98]	2025	physical realism	open	--
Physion [12]	2021	physical prediction from vision	open	--
Physion++ [131]	2023	physical prediction with online property inference	open	--
PhyWorldBench [57]	2025	physical realism in text-to-video	open	--
VideoPhy [7]	2024	physical commonsense in generated video	open	--
VideoPhy-2 [8]	2025	action-centric physical commonsense	open	--
<i>Counterfactual reasoning</i>				
CLEVRER [142]	2019	physical/causal video reasoning (counterfactual)	open	--
WorldPrediction [25]	2025	counterfactual action recognition	open	--
<i>Autonomous driving</i>				
Vista [51]	2024	driving world model	open	--
Bridges (prediction to action) (4)				
<i>Other</i>				
RoboWM-Bench [70]	2026	prediction executability	bridge	~
World-in-World [148]	2025	closed-loop task success of WMs	bridge	~
WorldArena [118]	2026	closed-loop WM arena	bridge	~
<i>Generation quality</i>				
WorldSimBench [109]	2024	video-to-action consistency	bridge	~